PASSES-HGV: Extending a Fixed-Grid Spectral Framework to 6-DOF Hypersonic Glide and Fractional Orbital Trajectories

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Abstract

This paper extends the fixed-grid spectral multi-physics formulation of the companion PASSES framework [5] from a one-dimensional beam to a two-dimensional lifting body, and from a single atmospheric arc to a trajectory spanning boost, exo-atmospheric coast, and hypersonic reentry. Three changes are required. First, the dense Chebyshev collocation operator, whose condition number grows as $\mathcal{O}(N^8)$ for fourth-order bending, is replaced by an ultraspherical spectral discretization [11, 16] in which the differential operators are banded and the conditioning is $\mathcal{O}(1)$; the bivariate extension is assembled as a Kronecker-structured generalized Sylvester system. Second, Euler–Bernoulli kinematics are replaced by a Mindlin–Reissner anisotropic plate formulation [9, 12], which admits the transverse shear deformation that dominates flutter behavior in thicker hull sections, at the cost of requiring three independent free-edge conditions rather than the classical Kirchhoff effective-shear pair. Third, the thermal model is split by material class: charring pyrolysis in the style of CMA [10] for phenolic acreage, and a single-temperature oxidative recession model for non-pyrolyzing refractory leading edges, both mapped to a fixed computational domain by a Landau transformation.

The aerothermal closure couples oblique shock-expansion pressure, Fay–Riddell stagnation convective heating [3] with a modified Lees distribution [6], and Tauber–Sutton radiative heating [15] directly to the deformed structural state. Trajectory generation is posed as a successive convexification problem [7, 14] with free virtual controls under an exact ℓ_1 penalty and a trust region governed by the actual-to-predicted cost reduction ratio. Navigation through plasma blackout uses an ionization-gated Extended Kalman Filter; we give the correct growth rates for the unaided covariance, which are t^3, t^4, and t^6 in the velocity-random-walk, accelerometer-bias, and gyro-bias-through-gravity channels respectively, rather than the quadratic growth sometimes assumed. Exo-atmospheric propagation retains the J_2 geopotential term [18].

This manuscript presents the formulation and a falsifiable verification plan. Numerical results are marked in place and are the subject of a companion experimental report.

Keywords: Ultraspherical Spectral Method, Mindlin–Reissner Plate Theory, Hypersonic Aerothermodynamics, Successive Convexification, Plasma Blackout, J_2 Perturbation, Applied Mathematics.

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Contents

Nomenclature

Symbol	Definition
Geometry and kinematics	
ξ, η	computational coordinates on the planform, $[-1, 1]^2$
$\mathbf{q}$	attitude quaternion, $[q_0, q_1, q_2, q_3]^\top$
$\boldsymbol{\omega}_B$	body angular rate, rad/s
$\mathbf{n}(\xi, \eta, t)$	outward unit normal of the deformed surface
$\hat{\mathbf{v}}_B$	unit freestream velocity in body axes
δ_c	local incidence angle, rad
α	vehicle angle of attack, rad
Structural	
w	transverse deflection, m
ϕ_x, ϕ_y	rotations of the cross-section normal, rad
h	plate thickness, m
$\mathbf{D}_{ij}$	anisotropic flexural rigidity components, N m
G_{xz}, G_{yz}	transverse shear moduli, Pa
κ_s^2	shear correction factor, 5/6
M_x, M_y, M_{xy}	bending and twisting moments, N
Q_x, Q_y	transverse shear forces, N/m
γ_{xz}, γ_{yz}	transverse shear strains
$\mathcal{D}_k$	ultraspherical k-th derivative operator
$\mathcal{S}_k$	ultraspherical basis conversion operator
$\otimes$	Kronecker product
Aerothermodynamics	
q_{dyn}	freestream dynamic pressure, Pa
C_p	pressure coefficient
$\dot{q}_{s,\mathrm{conv}}$	stagnation convective heat flux, W/m^2
$\dot{q}_{s,\mathrm{rad}}$	stagnation radiative heat flux, W/m^2
Pr, Le	Prandtl and Lewis numbers
β_{Le}	Lewis exponent, 0.52 equilibrium / 0.63 frozen
h_{0e}, h_w	total edge and wall enthalpy, J/kg
h_D	dissociation enthalpy, J/kg
ρ_e, μ_e	boundary-layer edge density and viscosity
ρ_w, μ_w	wall density and viscosity
R_{eff}	effective nose radius, m
$\dot{s}$	surface recession rate, m/s
ΔH_{abl}	effective ablation enthalpy, J/kg
GNC and orbital	
$\mathbf{A}_i, \mathbf{B}_i$	discretized linearized dynamics matrices
$\boldsymbol{\nu}_i$	virtual control vector
w_ν	exact-penalty weight
ρ_k	trust-region actual-to-predicted reduction ratio
$\Delta L^{(k)}$	predicted (convex) cost reduction
n_e	plasma electron number density, m^{-3}
ω_p	plasma angular frequency, rad/s
$\mathbf{P}(t)$	navigation state covariance
J_2	second zonal geopotential harmonic

Symbol	Definition
$\mu, R_{\oplus}$	Earth gravitational parameter and equatorial radius

1 Introduction

Hypersonic glide vehicles and fractional orbital systems present a coupled multi-physics problem that strains conventional computational aerospace methods along several axes simultaneously. The vehicle is a lifting body, so one-dimensional beam kinematics are insufficient. It flies at incidence angles and Mach numbers where aerodynamic loading and aerothermal heating are strongly coupled to the deformed shape, so aerodynamic look-up tables indexed on rigid-body attitude alone lose validity. And it traverses regimes — dense atmosphere, near-vacuum coast, reentry — whose governing balances differ by orders of magnitude, so a formulation tuned to one is typically unusable in another.

The companion paper [5] established a fixed-grid approach to the atmospheric problem: discretize each spatial domain once, hold the grid fixed, and advance the entire coupled system as a single system of ordinary differential equations. That paper's structural model is a one-dimensional Euler–Bernoulli beam discretized by dense Chebyshev collocation with null-space boundary projection. This paper extends the same principle to a two-dimensional lifting surface and a full-trajectory arc. Three extensions are required, each forced by a specific failure of the one-dimensional formulation.

Conditioning. The dense Chebyshev representation of a fourth-order operator has a condition number growing as $\mathcal{O}(N^8)$. In one dimension at modest N this is tolerable, and [5] manages it by null-space projection. In two dimensions the tensor-product operator inherits the same growth while the problem size grows as N^2, and the dense operator becomes both ill-conditioned and expensive. This paper adopts the ultraspherical spectral method of Olver and Townsend [11], in which differential operators are banded and the conditioning is $\mathcal{O}(1)$, with the bivariate construction following [16] and the resulting Kronecker-structured system solved by the blocked recursion of [1].

Kinematics. Euler–Bernoulli theory assumes cross-sections remain normal to the deformed axis, excluding transverse shear. For the thicker sections of a waverider hull, transverse shear is not a correction — it is the dominant contributor to the torsional modes that govern flutter. This paper adopts Mindlin–Reissner anisotropic plate theory [9, 12], in which the normal rotations ϕ_x, ϕ_y are independent fields rather than gradients of w.

Regime transitions. A trajectory spanning boost, coast, and reentry passes through conditions where the atmospheric terms vanish and the orbital terms dominate, and back. Section 7 describes how the framework handles this within a single integration.

1.1 Scope and relationship to the companion paper

This manuscript is a formulation paper: it presents the mathematical model, the discretization, and a falsifiable verification plan. Since the plan was written, a reference implementation has been built and the verification tasks executed; where a claim in this paper has been measured, the measured value is quoted inline and attributed, and where a claim proved wrong it has been corrected in place with the correction identified as such (see Remark 4.3). Detailed numerical results, figures, and the full experimental protocol remain the subject of a companion report. Claims not accompanied by a measurement are restricted to what follows analytically or by direct reference to prior work.

Readers should treat [5] as prerequisite for the fixed-grid rationale, the Landau transformation used for ablation, the adaptive filtering architecture, and the Monte Carlo dispersion estimators, none of which are re-derived here. Where this paper's choices differ from that one, the difference is flagged and justified rather than left for the reader to discover.

2 Related Work

Well-conditioned spectral methods. Chebyshev collocation [17] is the reference discretization for smooth problems on tensor-product domains, but its dense operators condition poorly at high differential order. Olver and Townsend [11] showed that recasting the problem in ultraspherical bases $C^{(\lambda)}$, with sparse conversion operators between them, yields banded differential operators and $\mathcal{O}(1)$ conditioning. Townsend and Olver [16] extended this to bivariate problems on rectangles, expressing a separable-rank-r operator as a generalized Sylvester equation. Chen and Kressner [1] give recursive blocked solvers for exactly this Kronecker structure. This paper's structural discretization is an application of that line of work to the Mindlin–Reissner system.

Plate theory. Reissner [12] and Mindlin [9] independently introduced first-order shear deformation theories that relax the Kirchhoff normality constraint. The resulting three-field formulation is standard in composite structures; its principal numerical hazard is shear locking as thickness tends to zero, which afflicts low-order finite elements severely. High-order spectral discretizations are substantially less susceptible, and Remark 5.3 states what remains.

Hypersonic aerothermal closure. Fay and Riddell [3] give the stagnation-point convective heating relation for dissociated air including finite-rate diffusion and surface catalysis; Sutton and Graves [13] give a simpler correlation for arbitrary gas mixtures that is widely used where the full Fay–Riddell inputs are unavailable. Lees [6] gives the laminar distribution of heating away from stagnation. Tauber and Sutton [15] give the radiative component, which becomes significant at high entry velocities. Charring ablation follows CMA [10] and FIAT [2].

Convex trajectory optimization. Mao, Szmuk, and Açıkmeşe [7] established the convergence properties of successive convexification with exact penalty and trust region; Szmuk, Reynolds, and Açıkmeşe [14] developed the real-time 6-DoF formulation with state-triggered constraints. The formulation in section 6.1 follows these directly.

Positioning. Each component above is established, and no individual model here is novel. The contribution claimed is the coupling: an ultraspherical Mindlin–Reissner structural state, an aerothermal closure evaluated on the deformed surface, and a convex guidance layer, all advanced as one system on grids that never move. Whether that coupling is worth its complexity relative to a conventional partitioned toolchain is an empirical question that section 8 is designed to answer.

3 The Expanded Global State

3.1 Attitude parameterization

Aggressive near-vertical dives and exo-atmospheric tumbling both pass through orientations where three-parameter attitude representations are singular. Attitude is therefore carried as a unit quaternion $\mathbf{q} = [q_0, q_1, q_2, q_3]^\top$ evolving by

$$\dot{\mathbf{q}} = \tfrac{1}{2}\,\boldsymbol{\Omega}(\boldsymbol{\omega}_B)\,\mathbf{q}, \qquad \boldsymbol{\Omega}(\boldsymbol{\omega}) = \begin{bmatrix} 0 & -\boldsymbol{\omega}^\top \\ \boldsymbol{\omega} & -[\boldsymbol{\omega}\times] \end{bmatrix}, \tag{3.1}$$

integrated concurrently with the structural state. As in [5], the unit-norm constraint is maintained by Baumgarte stabilization within the right-hand side rather than by post-step renormalization, so that the correction remains visible to the adaptive step-size controller.

3.2 Local incidence angle on the deformed surface

Because the waverider has significant planform curvature and deforms aeroelastically in flight, a single vehicle angle of attack α does not determine the local flow condition at a given surface station. The local incidence angle is defined pointwise from the deformed outward normal $\mathbf{n}(\xi, \eta, t)$ and the unit body-frame velocity $\hat{\mathbf{v}}_B$:

$$\sin \delta_c(\xi, \eta, t) = -\,\mathbf{n}(\xi, \eta, t) \cdot \hat{\mathbf{v}}_B. \tag{3.2}$$

Remark 3.1 (Sign convention)**.** The negative sign in eq. (3.2) is required and is easy to lose. With $\mathbf{n}$ the *outward* normal, a windward panel — one facing into the flow — has $\mathbf{n} \cdot \hat{\mathbf{v}}_B < 0$. Defining $\sin \delta_c$ without the negation makes windward panels register as leeward, silently inverting the entire pressure and heating distribution. The convention adopted here is $\delta_c > 0$ windward, $\delta_c \leq 0$ leeward.

The normal $\mathbf{n}$ is computed from the deformed surface parameterization, so it depends on the structural state (w, ϕ_x, ϕ_y); this is the coupling that makes the aerodynamic load a function of the deformation rather than of rigid-body attitude alone.

3.3 Aerodynamic forcing on the spectral grid

Modified Newtonian pressure alone is inadequate for slender waveriders at moderate incidence: it assigns zero pressure coefficient to all leeward surfaces, discarding leeward suction, and it takes no account of viscous interaction. PASSES-HGV therefore uses a blended closure. On windward panels ($\delta_c > 0$) the local pressure coefficient follows oblique shock-expansion theory, reducing to modified Newtonian

$$C_p = C_{p,\max} \sin^2 \delta_c, \qquad C_{p,\max} = \frac{2}{\gamma M_\infty^2}\left(\frac{p_{02}}{p_\infty} - 1\right), \tag{3.3}$$

where the pressure ratio across the normal shock at the stagnation point is given by the Rayleigh–Pitot relation. On leeward panels ($\delta_c \leq 0$), Prandtl–Meyer expansion is applied until the vacuum limit $C_p = -2/(\gamma M_\infty^2)$ is reached, beyond which the surface is treated as being at vacuum pressure.

The distributed forcing vector on the collocation grid is then

$$\mathbf{Q}_{\text{aero}} = q_{dyn}\, \mathbf{C}_p, \qquad q_{dyn} = \tfrac{1}{2}\rho_\infty V_\infty^2. \tag{3.4}$$

Remark 3.2. Equation (3.3) and its leeward counterpart together are C^0 but not C^1 at $\delta_c = 0$, since the windward branch has zero slope there while the expansion branch does not. This is a genuine violation of the global smoothness the framework otherwise maintains, and it occurs precisely at the shoulder line where panels transition between branches. The two branches are blended over a narrow band $|\delta_c| < \delta_{\text{blend}}$ by a C^2 smoothstep, with δ_{blend} chosen small relative to the incidence variation across a collocation cell. This is a numerical expedient, not physics, and its effect on integrated loads is verification task V4 in section 8.

4 Hypersonic Aerothermodynamics

4.1 Stagnation-point convective heating

The convective heat flux at the stagnation point follows Fay and Riddell [3], which accounts for boundary-layer equilibrium chemistry and the degree of surface catalytic recombination:

$$\dot{q}_{s,\text{conv}} = 0.763\ \overset{-0.6}{\Pr}\,(\rho_e\mu_e)^{0.4}(\rho_w\mu_w)^{0.1}\sqrt{\left(\frac{\mathrm{d}u_e}{\mathrm{d}x}\right)_s}\,(h_{0e} - h_w)\left[1 + \left(Le^{\beta_{Le}} - 1\right)\frac{h_D}{h_{0e}}\right]. \tag{4.1}$$

The terms require specification, since eq. (4.1) is frequently quoted without them:

- $\Pr \approx 0.71$ for air; $Le \approx 1.4$.
- $\beta_{Le} = 0.52$ for an equilibrium boundary layer, $\beta_{Le} = 0.63$ for a frozen boundary layer with a fully catalytic wall. The choice matters: the bracketed factor differs by several percent between them, and by considerably more for a non-catalytic wall, for which the bracket reduces toward unity.
- h_D is the enthalpy of dissociation of the free-stream mixture, evaluated at edge conditions.
- The stagnation velocity gradient is supplied by the modified Newtonian estimate

$$\left(\frac{\mathrm{d}u_e}{\mathrm{d}x}\right)_s = \frac{1}{R_{\text{eff}}}\sqrt{\frac{2(p_s - p_\infty)}{\rho_s}}. \tag{4.2}$$

Equation (4.2) makes the coupling to ablation explicit: recession increases R_{eff}, which reduces the velocity gradient, which reduces $\dot{q}_{s,\text{conv}}$ as $R_{\text{eff}}^{-1/2}$. Nose blunting is self-limiting, and capturing that requires the recession model of section 4.4 to feed back into eq. (4.2) within the same time step.

Remark 4.1 (Where the constants in eq. (4.1) actually come from)**.** Implementing eq. (4.1) against Fay and Riddell's original paper rather than against its restatements produced two findings worth recording.

First, the split exponents are *derivable* rather than conventional. The source's Eq. (45) writes the flux as $q = (Nu/\sqrt{Re})\sqrt{\rho_w\mu_w(\mathrm{d}u_e/\mathrm{d}x)_s}\,(h_s - h_w)/\Pr$, and its Eq. (62) correlates $Nu/\sqrt{Re} = 0.67\,(\rho_s\mu_s/\rho_w\mu_w)^{0.4}\{1 + (Le^{0.52} - 1)h_D/h_s\}$ — a *ratio* across the boundary layer, since the parameter "was found to depend only upon the total variation in $\rho\mu$." Substituting the second into the first gives the 0.4 from the ratio and a residual $0.5 - 0.4 = 0.1$ from the $\sqrt{\rho_w\mu_w}$, which is exactly the $(\rho_e\mu_e)^{0.4}(\rho_w\mu_w)^{0.1}$ of eq. (4.1). This matters practically: the archived scan of Eq. (63) is too degraded to read the exponents off directly, and the assignment can be recovered instead.

Second, and less comfortably, that same substitution shows every constant in the correlation descending from *one* fitted number carried at two significant figures — the 0.67 of Eq. (58)/(62). The 0.94 printed in Eq. (63) is $0.67/0.71 = 0.9437$ rounded; the $\Pr^{-0.6}$ of the footnote is the correlation's own $\Pr^{0.4}$ scaling set against the $\Pr^{-1}$ of Eq. (45). Backing the eq. (4.1) coefficient out of the paper's three statements therefore gives three answers: $0.67\,\Pr^{-0.4} = 0.7684$ from Eq. (62), $0.94\,\Pr^{0.6} = 0.7654$ from Eq. (63), and 0.76 from the footnote as printed — a band of 1.1%. The 0.763 we write above, and which is in general circulation, appears nowhere in the source, though it sits inside that band and is as defensible as any of them.

We are not claiming an error. The spread is around one percent in $\dot{q}_s$, well inside the uncertainty of the edge properties that feed the correlation. We record it because a heat-flux coefficient quoted to three or four significant figures implies a traceability that the source does not support at more than two, and because a reader reproducing our results from the original paper would otherwise find a mismatch and reasonably assume an implementation error. Our implementation names all four constants and defaults to 0.763 for continuity with the literature rather than fidelity to the source.

Remark 4.2. Where the inputs to eq. (4.1) are unavailable — typically in early trajectory screening, before a boundary-layer edge solution exists — the framework falls back to the Sutton–Graves correlation [13], $\dot{q}_s = k_{SG}\sqrt{\rho_\infty/R_{\text{eff}}}\,V_\infty^3$ with $k_{SG} = 1.7415 \times 10^{-4}$ in SI units for air. This is a correlation, not a theory, and results obtained with it should not be reported as Fay–Riddell results. The distinction is noted because conflating the two is a common error.

4.2 Radiative heating

At the entry velocities relevant to fractional orbital return, shock-layer radiation is not negligible relative to convection. The Tauber–Sutton correlation [15] gives

$$\dot{q}_{s,\mathrm{rad}} = C\, R_{\mathrm{eff}}^{a}\, \rho_{\infty}^{b}\, f(V_{\infty}), \tag{4.3}$$

where, for air,

$$C = 4.736 \times 10^{4}, \qquad a = 1.072 \times 10^{6}\, V_{\infty}^{-1.88} \rho_{\infty}^{-0.325}, \qquad b = 1.22, \tag{4.4}$$

with a additionally capped by nose radius — $a \leq 0.6$ for $1 \leq R_{\mathrm{eff}} \leq 2$ m and $a \leq 0.5$ for $2 < R_{\mathrm{eff}} \leq 3$ m — and subject throughout to the source's requirement that $a < 1$. The velocity function f is tabulated and rises steeply, effectively as V_{∞}^{8} to V_{∞}^{10} over the correlation's validity range.

Remark 4.3 (Three properties of eq. (4.3) that are easy to lose)**.** An earlier draft of this paper described the nose-radius exponent as $a \approx 1.0$. That is incorrect: a is a function of both velocity and density, it is capped by nose radius, and the source states explicitly that $a < 1$ must always be met. Over the correlation's own validity envelope it runs approximately 0.25 to 0.6. The qualitative claim below — that blunting *increases* radiative heating — is unaffected, since $a > 0$; the quantitative exponent is.

Second, eq. (4.3) returns $\dot{q}_r$ in W/cm^2, not W/m^2. A framework carrying SI units throughout requires a factor of 10^4, and omitting it produces a four-order-of-magnitude error in a quantity that still resembles a heat flux.

Third, the tabulated f_E extends from 9 km/s, but eq. (4.3)–eq. (4.4) are stated to apply only for $10 \leq V_{\infty} \leq 16$ km/s, $6.66 \times 10^{-5} \leq \rho_{\infty} \leq 6.31 \times 10^{-4}$ kg/m^3, and $0.3 \leq R_{\mathrm{eff}} \leq 3$ m. Interpolating the table below 10 km/s is legitimate; evaluating the correlation there is extrapolation of a fit beyond where its authors validated it, and the implementation refuses it unless asked explicitly.

Note that the R_{eff} dependence in eq. (4.3) is of *opposite sign* to that in eq. (4.2): blunting reduces convective heating but increases radiative heating. The optimal nose radius is therefore a genuine trade rather than a monotone preference, and a framework that models only convection will systematically favor over-blunted geometries. Evaluated on the published correlation at $V_{\infty} = 12$ km/s and $\rho_{\infty} = 3 \times 10^{-4}$ kg/m^3, the total stagnation heating has an interior minimum near $R_{\mathrm{eff}} \approx 0.8$ m — inside the source's own nose-radius envelope, and at a condition where radiation exceeds convection.

4.3 Heat flux distribution over the planform

Stagnation heating is mapped across the bivariate grid using the local incidence angle. For windward nodes a modified Lees distribution [6] governs the local flux:

$$\dot{q}_{\mathrm{total}}(\xi, \eta) = (\dot{q}_{s,\mathrm{conv}} + \dot{q}_{s,\mathrm{rad}}) \sin \delta_c(\xi, \eta) \left(\frac{R_{\mathrm{eff}}}{x(\xi)} \right)^{1/2}, \tag{4.5}$$

where $x(\xi)$ is the running length along the surface streamline from the stagnation point. Equation (4.5) is singular as $x \to 0$ and is therefore evaluated as $\dot{q}_s$ within the stagnation region $x < R_{\mathrm{eff}}$, matching continuously at $x = R_{\mathrm{eff}}$.

Remark 4.4. Equation (4.5) presumes laminar flow. Transition to turbulence raises local heating by a factor of three to five and is the single largest source of uncertainty in windward heating prediction. The framework carries a transition criterion based on momentum-thickness Reynolds number as a switchable model, and dispersion analyses should sample over transition location rather than fixing it, since the resulting uncertainty typically dominates the aerothermal error budget.

4.4 Surface recession on a fixed computational domain

The companion paper [5] argues against treating ablation as a single-temperature Stefan phase change, on the grounds that charring composites pyrolyze across an extended reaction zone and that a Stefan model omits both the pyrolysis gas enthalpy and the blowing reduction of convective heating. That argument stands and is not revisited here. It does not, however, apply uniformly across an HGV thermal protection system, and this paper therefore splits the model by material class rather than adopting one formulation everywhere.

Acreage TPS (charring). Phenolic-impregnated and carbon-phenolic acreage materials pyrolyze. These regions use the full two-phase charring formulation of [5], comprising Arrhenius decomposition kinetics, in-depth pyrolysis gas transport, and a surface energy balance with the blowing reduction factor $\phi = \ln(1 + 2\lambda B')/(2\lambda B')$. That model is not reproduced here.

Leading edges (non-pyrolyzing). Sharp leading edges and control surfaces on an HGV are typically refractory — carbon-carbon, or an ultra-high-temperature ceramic such as ZrB_2–SiC. These materials do not pyrolyze. They recede by surface oxidation and mechanical removal, with no in-depth reaction zone and no pyrolysis gas. For these regions a single-temperature moving-boundary energy balance is the physically correct model, not a simplification:

$$\rho_m \Delta H_{\mathrm{abl}}\, \dot{s}(\xi,\eta,t) = \dot{q}_{\mathrm{total}}(\xi,\eta,t) - \varepsilon\sigma_S T_w^4 - k_{\mathrm{mat}} \left.\frac{\partial T}{\partial n}\right|_{\mathrm{wall}}. \tag{4.6}$$

Remark 4.5 (Resolving the apparent inconsistency)**.** Equation (4.6) is the Stefan-type balance that [5] argues against for charring materials, and its appearance here is deliberate rather than a regression. The two models apply to different material classes. Applying eq. (4.6) to a charring phenolic would understate in-depth heating and overstate surface heat flux by omitting blowing; applying the charring model to a non-pyrolyzing ceramic would introduce decomposition kinetics for a reaction that does not occur. The framework selects per region from the material map.

In both cases the receding domain is mapped to a fixed computational domain by the Landau transformation of [5], so that recession appears as a grid-velocity advection term and no node is created, destroyed, or interpolated. The recession rate updates R_{eff} in eq. (4.2) and eq. (4.3) and the deformed normals $\mathbf{n}(\xi,\eta,t)$ in eq. (3.2), closing the aerothermal-structural loop.

5 Structural Model: Mindlin–Reissner Anisotropic Plates

5.1 The three-field structural state

A hypersonic lifting body is not governed by one-dimensional beam mechanics, and classical Kirchhoff–Love thin-plate theory — which constrains the cross-section normal to remain normal to the deformed midsurface — omits transverse shear deformation. In the thicker hull sections, transverse shear is the dominant mechanism of the torsional modes that set flutter margins. The structural state at each spatial coordinate is therefore the three-field vector

$$\mathbf{u}(\xi,\eta,t) = \begin{bmatrix} w \\ \phi_x \\ \phi_y \end{bmatrix}, \tag{5.1}$$

in which the normal rotations ϕ_x, ϕ_y are independent fields rather than gradients of w. The transverse shear strains are consequently nonzero:

$$\gamma_{xz} = \phi_x + \frac{\partial w}{\partial x}, \qquad \gamma_{yz} = \phi_y + \frac{\partial w}{\partial y}. \tag{5.2}$$

Kirchhoff–Love theory is recovered in the limit $\gamma_{xz} = \gamma_{yz} = 0$, i.e. $\phi_x = -\partial w/\partial x$.

5.2 Constitutive relations and governing equations

For an orthotropic laminate with principal axes aligned to the planform coordinates, the moment and shear resultants are

$$\begin{bmatrix} M_x \\ M_y \\ M_{xy} \end{bmatrix} = \begin{bmatrix} \mathbf{D}_{11} & \mathbf{D}_{12} & 0 \\ \mathbf{D}_{12} & \mathbf{D}_{22} & 0 \\ 0 & 0 & \mathbf{D}_{66} \end{bmatrix} \begin{bmatrix} \partial\phi_x/\partial x \\ \partial\phi_y/\partial y \\ \partial\phi_x/\partial y + \partial\phi_y/\partial x \end{bmatrix}, \tag{5.3}$$

$$Q_x = \kappa_s^2 G_{xz} h\,\gamma_{xz}, \qquad Q_y = \kappa_s^2 G_{yz} h\,\gamma_{yz}, \tag{5.4}$$

with shear correction factor $\kappa_s^2 = 5/6$ following Reissner [12]; Mindlin's $\pi^2/12$ is an alternative differing by under 2%. The rigidity components $\mathbf{D}_{ij}(\xi,\eta)$ form the spatially continuous anisotropic flexural rigidity tensor, constructed by hyperbolic blending across material interfaces so that they remain differentiable.

The governing equations are then

$$\rho h \frac{\partial^2 w}{\partial t^2} = \frac{\partial Q_x}{\partial x} + \frac{\partial Q_y}{\partial y} + q, \tag{5.5}$$

$$\frac{\rho h^3}{12}\frac{\partial^2 \phi_x}{\partial t^2} = \frac{\partial M_x}{\partial x} + \frac{\partial M_{xy}}{\partial y} - Q_x, \tag{5.6}$$

$$\frac{\rho h^3}{12}\frac{\partial^2 \phi_y}{\partial t^2} = \frac{\partial M_{xy}}{\partial x} + \frac{\partial M_y}{\partial y} - Q_y. \tag{5.7}$$

Note that eq. (5.5)–eq. (5.7) are second order in each field, in contrast to the fourth-order Kirchhoff equation. This is the structural reason the three-field formulation is attractive for spectral discretization: the highest derivative order is halved, and with it the conditioning penalty.

5.3 Free-edge boundary conditions

Applying the classical Kirchhoff effective-shear condition to a Mindlin–Reissner state over-constrains the perimeter and induces an artificial boundary layer. Because the three fields are independent, three independent conditions are required per free edge. Along a free edge normal to x:

$$M_x = 0, \qquad M_{xy} = 0, \qquad Q_x = 0, \tag{5.8}$$

which written out are

$$\mathbf{D}_{11}\frac{\partial \phi_x}{\partial x} + \mathbf{D}_{12}\frac{\partial \phi_y}{\partial y} = 0, \quad \mathbf{D}_{66}\left(\frac{\partial \phi_x}{\partial y} + \frac{\partial \phi_y}{\partial x}\right) = 0, \quad \kappa_s^2 G_{xz} h \left(\phi_x + \frac{\partial w}{\partial x}\right) = 0. \tag{5.9}$$

The third condition is evaluated independently of the geometric slope — this is exactly the freedom the three-field formulation provides, and collapsing it back onto $\partial w/\partial x$ reintroduces the Kirchhoff constraint the formulation exists to relax.

Remark 5.1 (Corner redundancy and how the conditions are imposed)**.** Imposing eq. (5.8) on all four edges produces a rank-deficient constraint set, and the deficiency is specific rather than incidental. The twist condition $M_{xy} = 0$ belongs to the free-edge triple of *both* edges meeting at a corner, so it is imposed twice there; $M_x \neq M_y$ and $Q_x \neq Q_y$ in general, so those are independent. The set is therefore deficient by exactly four, one per corner. A uniformly simply-supported perimeter, used here as a verification configuration because it admits a closed-form Mindlin solution, is deficient by twelve.

Because the deficiency is a property of the boundary configuration rather than of the discretization, the conditions are imposed by null-space projection — the same treatment the companion paper applies to the free-free beam [5] — which degrades gracefully under redundancy where row replacement does not. In implementation the rank is measured rather than assumed, so an inconsistent boundary assembly surfaces as an out-of-range deficiency instead of as quietly wrong frequencies.

Remark 5.2 (Free edges converge algebraically)**.** Free-edge plates carry weak stress singularities at the corners. In consequence the free-free spectrum converges *algebraically* in N, whereas the simply-supported spectrum on the same operator converges exponentially — measured here to 3.7×10^{-10} relative on the first ten modes at $N = 20$ against the closed-form Mindlin solution. This is a property of the boundary-value problem, not of the discretization, and it means a free-free frequency comparison at modest N should not be expected to reach the tolerance an analytic case reaches. Verification of the plate kernel therefore rests on the closed-form simply-supported reference and on manufactured solutions; the free-free comparison is reported alongside without a pass criterion attached.

Remark 5.3 (Shear locking)**.** As $h \to 0$ the shear stiffness in eq. (5.4) grows as h^{-1} relative to the bending stiffness, and a discretization unable to represent $\gamma_{xz} \to 0$ exactly develops spurious stiffness — shear locking. This is severe for low-order finite elements and motivates the reduced-integration and mixed formulations standard in that setting. High-order spectral discretizations are markedly less susceptible, because the polynomial space is rich enough to represent the constrained limit closely. "Less susceptible" is not "immune": locking behavior should be measured against thickness ratio rather than assumed absent, which is verification task V2 in section 8. Measured on the present discretization at $N = 16$ per direction, the fundamental free-free frequency parameter falls from 11.71 at $h/L = 0.2$ — genuine shear softening of a thick section — to a plateau of 13.307, and at $h/L = 10^{-3}$ differs from that plateau by +0.018%, against a 1% criterion. No locking is observed across three decades. What *is* paid as the section thins is conditioning: the shear-to-bending stiffness ratio grows as h^{-2}, and the rigid-body modes separate from the elastic spectrum by correspondingly fewer decades, which is what eventually limits the usable thickness range.

5.4 Ultraspherical discretization and Kronecker structure

Dense Chebyshev differentiation matrices condition as $\mathcal{O}(N^{2k})$ for a k-th order operator, giving $\mathcal{O}(N^8)$ for fourth-order bending. This is the same figure quoted in [5], and it is the reason that paper requires null-space projection.

The ultraspherical method [11] avoids the problem at its source. Representing the solution in the Chebyshev basis T_n but the k-th derivative in the ultraspherical basis $C^{(k)}$, the differentiation operator becomes diagonal with a single nonzero band,

$$\mathcal{D}_k = 2^{k-1}(k-1)! \begin{bmatrix} \underbrace{0 \ \cdots \ 0}_{k} & k & & \\ & & k+1 & \\ & & & \ddots \end{bmatrix}, \tag{5.10}$$

and basis conversion between $C^{(\lambda)}$ and $C^{(\lambda+1)}$ is effected by banded operators $\mathcal{S}_\lambda$. Variable coefficients enter as multiplication operators, which are banded whenever the coefficient has a rapidly decaying Chebyshev expansion — which the hyperbolically blended rigidity fields do by construction. The assembled operator is banded with $\mathcal{O}(1)$ condition number, and boundary conditions are appended as dense rows rather than substituted, preserving the bandedness of the interior.

For the bivariate problem on $[-1, 1]^2$, a separable operator of rank r is written

$$\mathcal{L} = \sum_{j=1}^{r} \mathbf{A}_j \otimes \mathbf{B}_j, \tag{5.11}$$

so that the discretized system is a generalized Sylvester equation in the matrix of bivariate coefficients [16]. The Mindlin–Reissner system eq. (5.5)–eq. (5.7) assembles as a 3×3 block operator, each block of the form eq. (5.11), and is solved by the recursive blocked algorithm of Chen and Kressner [1], which exploits the Kronecker structure rather than forming the $\mathcal{O}(N^4)$ dense matrix.

Remark 5.4. The $\mathcal{O}(1)$ conditioning claim is a property of the ultraspherical operator itself and is established in [11]. It does not automatically survive the addition of dense boundary rows, variable coefficients with slowly decaying expansions, or the block coupling of eq. (5.5)–eq. (5.7). We assert only what [11] establishes and measure the rest: the conditioning of the assembled Mindlin–Reissner block operator against N is verification task V1.

Measured, the assembled block operator is essentially $\mathcal{O}(1)$-conditioned: κ moves from 101 to 108 as N goes from 8 to 18 per direction, a fitted log–log slope of 0.09. The block coupling is therefore benign. Two qualifications matter. First, the result is *scaling-dependent*: measured with column equilibration alone the same operator appears to grow as $N^{1.6}$, because a block operator mixes entries carrying different physical units — transverse shear stiffness in N/m against bending rigidity in N·m — and a one-sided scaling leaves that imbalance in the matrix. Two-sided (Ruiz) equilibration removes it. A condition number quoted for a block operator without stating its scaling is not a well-defined quantity. Second, the boundary rows are *not* benign: the projected pencil actually solved grows far faster than the interior, which is precisely the caveat above and localizes where a preconditioner would have to act.

6 Guidance and Navigation

6.1 Successive convexification

Trajectory generation under thermal and structural constraints is posed as a non-convex optimal control problem and solved by successive convexification [7, 14]. At iteration k the nonlinear dynamics are linearized about the current reference trajectory and discretized, giving

$$\mathbf{x}_{i+1} = \mathbf{A}_i\mathbf{x}_i + \mathbf{B}_i\mathbf{u}_i + \mathbf{z}_i + \boldsymbol{\nu}_i, \qquad i = 1, \dots, N-1, \tag{6.1}$$

where $\boldsymbol{\nu}_i \in \mathbb{R}^{n_x}$ are virtual controls.

Remark 6.1 (Virtual controls are free, not sign-constrained)**.** The virtual controls in eq. (6.1) are *unconstrained* real vectors. Constraining them to be non-negative — as is sometimes done by analogy with slack variables in inequality constraints — is incorrect here and defeats their purpose. Virtual controls exist to absorb the artificial infeasibility introduced by linearizing non-convex dynamics, and that infeasibility has no preferred sign: the linearized trajectory may need correction in either direction along any state axis. A non-negativity constraint makes the subproblem infeasible in exactly the cases the virtual control was introduced to rescue, and it invalidates the exact-penalty argument of [7], which requires the penalty function to be exact on a neighborhood of zero in all directions.

Because $\boldsymbol{\nu}_i$ is free, it must be penalized to ensure it vanishes at convergence. The exact ℓ_1 penalty

$$J_\nu = w_\nu \sum_{i=1}^{N-1} \|\boldsymbol{\nu}_i\|_1 \tag{6.2}$$

is added to the objective. The ℓ_1 norm is chosen over ℓ_2 specifically because it is an *exact* penalty: for w_ν larger than a finite threshold related to the dual variables of the original problem, the minimizer of the penalized problem coincides with the constrained minimizer, so $\boldsymbol{\nu}_i = \mathbf{0}$ exactly rather than asymptotically. A quadratic penalty drives $\boldsymbol{\nu}_i$ to zero only as $w_\nu \to \infty$, degrading conditioning.

A trust region bounds the step, accepted or rejected on the ratio of actual to predicted cost reduction:

$$\rho_k = \frac{J\big(\mathbf{x}^{(k)}\big) - J\big(\mathbf{x}^{(k+1)}\big)}{\Delta L^{(k)}}, \qquad \Delta L^{(k)} = L\Big(\mathbf{x}^{(k)}\Big) - L\Big(\mathbf{x}^{(k+1)}\Big), \tag{6.3}$$

where J is the true nonlinear cost and L the convex model cost. The step is rejected and the radius contracted when $\rho_k < \rho_0$; accepted with contraction for $\rho_0 \le \rho_k < \rho_1$; accepted unchanged for $\rho_1 \le \rho_k < \rho_2$; and accepted with expansion for $\rho_k \ge \rho_2$. Note that $\Delta L^{(k)} > 0$ by construction whenever the convex subproblem makes progress, so eq. (6.3) is well posed; a nonpositive $\Delta L^{(k)}$ indicates subproblem stagnation and terminates the iteration.

6.2 Plasma blackout and ionization-gated estimation

During entry, the shock layer ionizes and the resulting plasma sheath attenuates or reflects radio-frequency signals, severing GNSS updates. The electron number density n_e is estimated from the Saha equation at post-shock conditions, giving the plasma angular frequency

$$\omega_p = \sqrt{\frac{n_e e^2}{m_e \varepsilon_0}}. \tag{6.4}$$

When $\omega_p \ge \omega_{\text{GNSS}}$ the signal is evanescent in the sheath and the measurement is unavailable. The EKF responds by zeroing the corresponding rows of the observation matrix, $\mathbf{H}_k \to \mathbf{0}$ on the GNSS block, so that the filter propagates on inertial measurements alone and the update step becomes a no-op for those channels.

Proposition 6.2 (Covariance growth during blackout)**.** *For an unaided strapdown inertial solution, the position error covariance during a blackout of duration t grows as*

$$\mathbf{P}_{rr}(t) \sim \underbrace{\tfrac{1}{3} q_a t^3}_{\textit{velocity random walk}} + \underbrace{\tfrac{1}{4}\sigma_{b_a}^2 t^4}_{\textit{accelerometer bias}} + \underbrace{\tfrac{1}{36} g^2 \sigma_{b_g}^2 t^6}_{\textit{gyro bias via gravity projection}}, \tag{6.5}$$

where q_a is the accelerometer white-noise PSD, $\sigma_{b_a}^2$ the accelerometer bias variance, and $\sigma_{b_g}^2$ the gyro bias variance.

Proof. Integrating white acceleration noise twice gives position variance $q_a t^3/3$. A constant accelerometer bias b_a produces position error $\frac{1}{2} b_a t^2$, whose variance is $\frac{1}{4}\sigma_{b_a}^2 t^4$. A constant gyro bias b_g produces attitude error $b_g t$, which misprojects the gravity vector by $g\, b_g t$, which integrates twice to position error $\frac{1}{6} g b_g t^3$, whose variance is $\frac{1}{36} g^2 \sigma_{b_g}^2 t^6$. □

Remark 6.3. Proposition 6.2 corrects a claim that is easy to make loosely. The covariance does not grow quadratically during blackout. Quadratic is the growth of the *position error* from an accelerometer bias; the corresponding *covariance* grows as t^4, and the gyro-bias channel grows as t^6 and dominates for blackouts beyond a few tens of seconds. The distinction is operationally significant: a guidance layer sizing its pull-up trigger on a quadratic model will under-predict the error growth badly at the durations that matter.

The expanding covariance is exposed to the SCvx layer of section 6.1 as a state-dependent cost, so that the optimizer can trade trajectory efficiency against navigation uncertainty — for instance by commanding a pull-up that reduces n_e enough to reacquire GNSS. Whether this trade is exercised usefully, or merely produces chattering at the blackout boundary, is validation task V6.

7 Orbital Mechanics and the Coast Phase

7.1 J_2-perturbed propagation

Over a fractional orbit, a spherical gravity model accumulates unacceptable secular error. The dominant correction is the J_2 zonal harmonic accounting for Earth oblateness [18]. In Earth-Centered Inertial Cartesian coordinates with $r = \|\mathbf{r}\|$,

$$\mathbf{g}(\mathbf{r}) = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{g}_{J_2}(\mathbf{r}), \tag{7.1}$$

$$\mathbf{g}_{J_2}(\mathbf{r}) = -\frac{3J_2\mu R_\oplus^2}{2r^5}\begin{bmatrix} x\left(1 - 5z^2/r^2\right) \\ y\left(1 - 5z^2/r^2\right) \\ z\left(3 - 5z^2/r^2\right)\end{bmatrix}, \tag{7.2}$$

with $J_2 = 1.08263 \times 10^{-3}$, $\mu = 3.986004418 \times 10^{14}$ m^3/s^2, and $R_\oplus = 6378137$ m. The secular effects are nodal regression and apsidal precession; over a fractional orbit the accumulated cross-range difference relative to a spherical model is of order kilometers, which is large compared with any meaningful terminal accuracy requirement.

Remark 7.1. Higher harmonics, luni-solar third-body terms, and solar radiation pressure are omitted. For a coast of under one orbital period this is defensible — J_2 exceeds the next zonal term by roughly three orders of magnitude — but the omission should be re-examined before the framework is applied to multi-revolution cases. Atmospheric drag during the coast is retained where the density model is non-negligible, since perigee altitude for a fractional orbital profile may be low enough for drag to be significant.

7.2 Regime transition within a single integration

The framework advances boost, coast, and reentry within one integration rather than handing off between phase-specific propagators. Handoffs are a known source of discontinuity: the state must be reinterpolated at the boundary, and the integrator restarts with no history.

The mechanism is that the atmospheric terms are not switched off at the Kármán line but decay smoothly with the density model, so that above roughly 100 km the aerodynamic and aerothermal contributions to the right-hand side become numerically negligible without any branch being taken. The structural state continues to be integrated during coast, where it rings down under whatever damping the model provides.

Remark 7.2 (Cost of the single-integration choice)**.** This choice is not free, and the trade should be stated honestly rather than presented as a pure gain. Continuing to integrate the full structural state through a coast phase in which it carries no meaningful energy means the step-size controller remains subject to the structural stability constraint — the $\mathcal{O}(N^{-4})$ bound of [5] — during a phase whose rigid-body dynamics would otherwise permit steps orders of magnitude larger. A pragmatic mitigation is to freeze the structural block once its modal energy falls below a threshold and reactivate it on atmospheric re-entry, which reintroduces a switch but confines it to a regime where the frozen state is provably quiescent. Which approach is preferable is verification task V5. Measured over a 300 s coast carrying a 100 rad/s structural mode, freezing the block once its modal energy has decayed to 10^{-6} of its initial value costs 15× fewer right-hand-side evaluations and 15× less wall-clock time, with both strategies conserving specific orbital energy to better than 10^{-8} per orbit. The measurement settles the trade in favor of freezing for coasts of this length, and the margin grows with coast duration because the ring-down time is fixed while the coast is not. The switch is defensible only because the freeze condition is checked rather than assumed.

7.3 Midcourse trajectory correction

A coast arc that is merely propagated accurately is not yet a trajectory that arrives anywhere. Injection errors, and the unmodeled accelerations the propagation deliberately omits, accumulate into a terminal miss that must be removed by one or more impulsive corrections. The targeting primitive is Lambert's problem: given the current position, an aimpoint, and the time remaining, find the conic joining them. We solve it in Izzo's formulation [4], whose independent variable renders the time of flight monotonic and so removes both the bracketing and the case analysis the classical universal-variable method requires near parabolic and half-revolution transfers.

Two things about applying it are worth recording, because both were discovered by measurement rather than anticipated.

The first is that Lambert's answer is not the maneuver. Lambert is a two-body solve, while the arc actually flown carries J_2; over a half-hour coast the resulting terminal miss is kilometres. That is negligible as a fraction of range and far too large as a correction residual. Lambert is therefore used only to seed a Newton iteration on the terminal-position sensitivity $\partial \mathbf{r}_f / \partial \mathbf{v}_0$, evaluated by central differences of the propagator itself so that it is guaranteed consistent with the dynamics being flown. Two passes bring the true miss below a metre. The seed set must also include the vehicle's current velocity, not merely Lambert's two branches: late in an arc the vehicle is nearly collinear with its own aimpoint, a geometry in which the transfer plane is barely determined and both Lambert branches can be nonsense — one case in our sweep returned a 127 km/s "correction".

The second concerns when to burn, and it corrects an intuition we held going in. Correcting a position error δr with time-to-go t_{go} costs $|\Delta \mathbf{v}| \sim |\delta r|/t_{go}$, which we confirm: the product of maneuver magnitude and time-to-go is constant along the arc to within about ten percent, and equals the uncorrected miss distance. Fuel therefore argues unambiguously for burning early. We had expected execution error to argue the other way, on the reasoning that a large early burn injects a correspondingly large new error. It does inject one — but that error does not penalise burning early, because it is *independent of when the burn occurs.* Execution error scales with the maneuver, the maneuver scales as $1/t_{go}$, and the terminal sensitivity to a velocity change scales as t_{go}; the two cancel. Isolating the effect by zeroing the navigation covariance, the execution-induced one-sigma miss moves from 97.6 m to 91.6 m across a nineteen-fold range of time-to-go over which the maneuver itself grows seventeen-fold.

What does favour a late burn is knowledge. A correction computed from an estimate can only remove the error the filter has observed; the remainder survives the maneuver untouched. As the navigation covariance contracts along the arc that residual contracts with it, and nothing pushes back — we find the expected miss falling monotonically to the end of the arc even at a pointing error of 100 mrad. The consequence is that midcourse burn timing is not a scalar optimisation with an interior optimum, as it is often presented, but a genuinely two-objective trade between propellant and terminal accuracy. The implementation reports the two separately rather than combining them, since the exchange rate between them is a mission decision and not a property of the dynamics.

7.4 Dispensing multiple vehicles from a post-boost bus

A bus carrying n independently targeted vehicles must place each on a ballistic arc reaching its own aimpoint. Each release is the targeting problem of section 7.3; what only appears once there is more than one release is the interaction between them. Three measured results are worth recording, and two of them contradict the conventional statement of the problem.

Ordering dominates the maneuvers. The assignment of aimpoints to release slots changes total Δv far more than any refinement of the individual maneuvers does. For four aimpoints spread roughly ±100 km downrange and ±60 km crossrange about a common impact point, enumerating all 24 orderings gives a spread of 85% between cheapest and dearest — 405 against 748 m/s — and the natural, index-order sequence is not merely suboptimal but is the *worst* of the twenty-four. Since each evaluation requires a full targeting solve per vehicle, exhaustive search is affordable only for small n; above that we use a 2-opt exchange and report the result as a local optimum rather than as an optimum.

The footprint anisotropy belongs to the timing freedom, not to the geometry. It is usually said that downrange separation is cheap and crossrange separation expensive. Under a *fixed* arrival epoch this is not so. Displacing an aimpoint 50 km from the nominal impact point at a release 600 s into a 2400 s arc costs 35.1 m/s downrange against 40.4 m/s crossrange — a 15% difference, not a qualitative one. The familiar asymmetry appears only when the arrival epoch is allowed to absorb the displacement: permitting arrival to slip by 10 s reduces the downrange cost to 16.1 m/s, a 54% saving, while leaving the crossrange and radial costs *exactly* unchanged with their optima at zero slip. A plane change cannot be bought with time; a downrange displacement can. The distinction matters because a mission that specifies a common arrival epoch has, without necessarily intending to, priced downrange separation as though it were crossrange.

Accumulated error and terminal dispersion are not the same ordering. Every bus maneuver is imperfect and the next vehicle inherits it, so the bus covariance grows monotonically through the sequence — each maneuver contributes a positive-semidefinite block and none is ever removed. The natural conclusion, that the last vehicle released is the least accurate, is false. A later release also has a shorter remaining flight, and terminal miss scales with that flight time through $\partial \mathbf{r}_f / \partial \mathbf{v}_0$, so the inherited error has less opportunity to grow into a miss. Over a three-vehicle sequence we measure one-sigma dispersions of 1482, 1983 and 1744 m: rising, then falling. Which vehicle needs the accuracy budget is a result of the schedule and cannot be read off the release order.

8 Verification and Validation Plan

This section states what is measured, against what reference, and what constitutes failure. The plan was written before any implementation existed, so it is falsifiable in advance rather than reported selectively afterward. Tasks V1–V7 have since been executed against the reference implementation; V8 is executed for the radiative correlation and its implementation properties, with the convective reference-case comparison outstanding. Measured values appear in the relevant sections above and in the companion repository's verification reports.

ID	Target	Method and reference	Failure criterion
V1	Ultraspherical operator	κ of the assembled Mindlin–Reissner block operator versus N; comparison against dense Chebyshev collocation	κ growing faster than $\mathcal{O}(N)$
V2	Shear locking	Free-free plate frequencies versus thickness ratio h/L across three decades; comparison against thin-plate Kirchhoff limit	spurious stiffening above 1% at $h/L = 10^{-3}$

ID	Target	Method and reference	Failure criterion
V3	Plate dynamics	Natural frequencies of an isotropic free-free square plate against published Rayleigh–Ritz values; method of manufactured solutions on eq. (5.5)–eq. (5.7)	relative frequency error $> 10^{-5}$ for the first ten modes
V4	Aerodynamic blending	Integrated normal force and pitching moment versus δ_{blend}; sensitivity of trim solution	trim incidence shifting by more than 0.1° across the blend-width sweep
V5	Regime transition	Coast-phase step size and wall-clock, single-integration versus frozen-structure; energy conservation over the coast	secular energy drift exceeding 10^{-8} per orbit
V6	Blackout navigation	Covariance growth against eq. (6.5); SCvx pull-up trigger behavior at the blackout boundary	measured growth not matching eq. (6.5) exponents; chattering at the boundary
V7	SCvx convergence	Iteration count and virtual control norm at convergence versus w_ν; verification that $\boldsymbol{\nu}_i \to \mathbf{0}$ exactly	$\|\boldsymbol{\nu}_i\|$ not reaching zero for finite w_ν
V8	Aerothermal	Stagnation heating against Fay–Riddell reference cases; recession against a FIAT [2] comparison	heating disagreement $>$ 5% on reference conditions

Status. V1–V7 executed and passing against the criteria as stated. V8 is partially complete. The radiative correlation is verified against the published source (Remark 4.3) and the implementation properties hold, but the 5% criterion is stated against published Fay–Riddell reference conditions, which require tabulated equilibrium-air properties not yet transcribed. The recession half of V8 — shared with Paper I's V4 — is now executed against measured arc-jet recession rather than against a code: seven conditions of Milos and Chen [8] spanning 107–1102 W/cm^2 and 2.3–84.4 kPa, at 4% median absolute error against a 27% experimental scatter. Full numerical detail is deferred to the companion experimental manuscript.

9 Limitations

1. **No results.** This is a formulation paper. Conditioning, locking, convergence, and throughput claims are arguments from structure or citations to prior work, not measurements of this implementation.

2. **Engineering-level aerodynamics.** Shock-expansion and modified Newtonian closures are inviscid and local. They do not represent shock–boundary-layer interaction, separated flow, base flow, or shock–shock interference at control surface junctions — the last being a principal driver of localized heating peaks on real vehicles. Numerical smoothness in the structural and thermal solvers does not compensate for error entering through this closure.

3. **Laminar heating distribution.** Equation (4.5) assumes laminar flow; transition is modeled by a switchable correlation with substantial uncertainty, which typically dominates the aerothermal error budget.

4. **First-order shear deformation.** Mindlin–Reissner theory assumes a constant transverse shear distribution through the thickness, corrected by κ_s^2. For thick sandwich construction with soft cores, higher-order or layerwise theories are required.

5. **Linear structural response.** Equation (5.5)–eq. (5.7) are linear. Geometric nonlinearity under large deflection and material nonlinearity under thermal softening are outside scope, as is aerothermoelastic buckling.

6. **Rectangular computational domain.** The tensor-product ultraspherical construction requires a topologically rectangular parameterization of the planform. Genuinely multiply-connected geometries, or planforms with re-entrant corners, require domain decomposition that is not addressed here.

7. **Constant plate rigidities in the reference implementation.** Equation (5.3) admits spatially varying $\mathbf{D}_{ij}(\xi, \eta)$, and the assembly is written to accept them, but the implementation exercised here uses constant coefficients. Genuinely bivariate coefficient fields require a low-rank approximation of each coefficient so that the Kronecker structure of eq. (5.11) survives; separable fields are a direct extension, general ones are not.

8. **Boundary conditioning, not operator conditioning, is the limiter.** The measurements reported in section 5.4 show the banded interior behaving as advertised while the bordered system does not. Any claim about the scalability of this discretization to large N is a claim about the boundary treatment, and that has not been addressed by preconditioning here.

9. **Equilibrium chemistry.** Equation (4.1) and the Saha estimate of eq. (6.4) assume equilibrium. At high altitude and high velocity the shock layer is chemically frozen or in nonequilibrium, and both the heating and the electron density will be misestimated.

10. **Simplified gravity and environment.** As stated in section 7.1, only J_2 is retained.

10 Conclusion

This paper extended the fixed-grid spectral formulation of [5] to two-dimensional lifting-body structures and full-trajectory arcs. The dense Chebyshev operator was replaced by an ultraspherical discretization with banded operators and Kronecker-structured bivariate assembly; Euler–Bernoulli kinematics were replaced by a three-field Mindlin–Reissner formulation with three independent free-edge conditions; and the thermal model was split by material class, with charring pyrolysis for phenolic acreage and single-temperature oxidative recession for non-pyrolyzing refractory leading edges, resolving what would otherwise be an inconsistency with the companion paper.

Several points were corrected relative to a naive treatment. Remark 3.1 fixes a sign convention in the local incidence angle whose loss silently inverts the entire load distribution. Remark 6.1 establishes that SCvx virtual controls must be free rather than sign-constrained, and why the ℓ_1 penalty is exact where a quadratic penalty is not. Proposition 6.2 gives the correct t^3, t^4, and t^6 covariance growth rates during plasma blackout in place of an assumed quadratic. Remark 7.2 states the cost of single-integration regime transition rather than presenting it as free.

What remains is measurement. Section 8 states in advance what will be measured and what would falsify the design.

Code and Data Availability

The manuscript sources, bibliographies, and citation audit for this work are available at `https://github.com/ethank5149/Parameterized-AeroStructural-State-Estimation-Sandbox`. The implementation, container specifications, and figure-generating scripts will be released under the MIT license in the same repository. **[PENDING: archive a tagged release with a Zenodo DOI at submission]**

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Appendix A Ultraspherical Operators

The ultraspherical (Gegenbauer) polynomials $C_n^{(\lambda)}$ satisfy the derivative relation

$$\frac{\mathrm{d}}{\mathrm{d}x} C_n^{(\lambda)}(x) = 2\lambda\, C_{n-1}^{(\lambda+1)}(x), \tag{A.1}$$

with the Chebyshev polynomials recovered as the limit $T_n = \lim_{\lambda\to 0} \frac{n}{2\lambda} C_n^{(\lambda)}$ for $n \geq 1$. Equation (A.1) is what makes $\mathcal{D}_k$ in eq. (5.10) sparse: differentiating raises the basis index by one and shifts the degree by one, so the operator has exactly one nonzero diagonal, in contrast to the dense collocation derivative.

Conversion between adjacent bases follows from

$$C_n^{(\lambda)} = \frac{\lambda}{\lambda + n} \left(C_n^{(\lambda+1)} - C_{n-2}^{(\lambda+1)} \right), \tag{A.2}$$

giving tridiagonal conversion operators $\mathcal{S}_\lambda$. A k-th order differential operator with variable coefficients is assembled as

$$\mathcal{L} = \sum_{j=0}^{k} \mathcal{M}_j^{(k)}\, \mathcal{S}_{k-1} \cdots \mathcal{S}_j\, \mathcal{D}_j, \tag{A.3}$$

where $\mathcal{M}_j^{(k)}$ is the multiplication operator for the j-th coefficient function, expressed in the $C^{(k)}$ basis. Bandwidth of $\mathcal{M}_j^{(k)}$ equals the truncation length of the coefficient's Chebyshev expansion, so slowly converging material property fields degrade the sparsity — which is the practical reason for constructing the rigidity fields by hyperbolic blending rather than by piecewise assignment.

Appendix B Quaternion Kinematics and Frame Conventions

The attitude quaternion $\mathbf{q}_{E2B}$ maps ECI to body axes, with scalar part first, $\mathbf{q} = [q_0, \boldsymbol{q}_{1:3}]^\top$, and the Hamilton (not JPL) product convention. The direction cosine matrix is

$$\mathbf{C}_E^B = \left(q_0^2 - \boldsymbol{q}_{1:3}^\top \boldsymbol{q}_{1:3} \right) \mathbf{I} + 2\, \boldsymbol{q}_{1:3} \boldsymbol{q}_{1:3}^\top - 2 q_0 \left[\boldsymbol{q}_{1:3} \times \right]. \tag{B.1}$$

Mixing Hamilton and JPL conventions produces a transpose error that is easy to introduce and hard to detect, since both yield valid rotation matrices; the convention is stated explicitly here for that reason.